CHAPTER 4:
Introduction to Client-Side Scripting (JavaScript)

Prepared for:

CSC264 - Introduction to Web and Mobile Application

OVERVIEW OF THIS CHAPTER



In this chapter, you will learn about:

- What is JavaScript
- Variables & Data Types
- Comments
- Output Operations
- Escape Sequences
- String-Handling

INTRODUCTION

JavaScript:

- Is a language similar to ordinary programming language

WHY?

Has variable declaration, branching, looping, array, and many more

- Basically, JavaScript is called a **scripting language**

WHY?

A JavaScript program does not need to be compiled, instead it is **interpreted** by a browser

INTRODUCTION

JavaScript:

- Is also known as **Client-Side Scripting**



WHY?

The scripts involved will be processed by the browser (client), not the server

- Will provide better performance and faster response time:
  Because it doesn't need to contact the sever to do any processing

- Allow web pages to be dynamic and interactive:
  By reacting to user events

INTRODUCTION

Client-Side Script vs Server Side Script

Client side script	Server side script
The action takes on the user's (client's) computer that is used to provide **user functionality** without needing to access a web server	The action takes place on a web server, as it is used to is used to interact with permanent storage like databases or files
Executed at browser	Executed at server
E.g.: JavaScript	E.g.: PHP

INTRODUCTION



Why do you need to learn JavaScript?

- Enables the developer to incorporate some programming features inside the website
- Will make the web page lively and responsive to changes from the browsers or the users

JAVASCRIPT

JavaScript:

- To write JavaScript, use the script element:

```
<script>
      .....
</script>
```

When the browser finds a <script> tag:
It will switch into Javascript mode and executes the code inside

- **ALL** JavaScript scripts will be embedded in HTML documents, either by:

1. **Including directly inside a HTML document**
   - The <script> tag can be placed anywhere in the web page depending on the suitability, but the most suitable place are:
     - i. **Inside the <head> section of the document:**
       Especially if it contains functions that are called throughout the page
     - ii. **At the <body> of the document:**
       The <script> tag can be used multiple times along the <body>

JAVASCRIPT

JavaScript:

- All JavaScript scripts will be embedded in HTML documents, either by:

2. **Linking to a Javascript document as an external file:**

   - The external JavaScript file will have an extension **.js**
   - To use the external JavaScript, the **src attribute** can be used

```
<script src="myScript.js"></script>
```

JAVASCRIPT: VARIABLES

Variables:

- Unlike HTML, JavaScript is case-sensitive
- JavaScript's variable can hold a value of **any data type**



  - You don't have to specify what type of value will be stored during variable declaration
  - The value type of a variable can change during the execution of a program and JavaScript takes care of it automatically

JAVASCRIPT: VARIABLES

Variables:

- To declare a variable in JavaScript, the following syntax isused:

```
var variableName;
```

Notice that the data type is not specified

- Each variable created must follow the following rules:
  - Begin with a letter or underscore ONLY
  - Consists of either letters, underscores,and numbers
  - Do not use any reserved keyword as variable name

- For example:

```
var fullName;
var age;
var  _gender;
var drink_option;
```

JAVASCRIPT: DATA TYPES

Data Types:

- JavaScript allows you to work with three primitive data types:

1. **Numbers:**
   - In JavaScript, there is no distinction between integer values and floating-point values:
     All numbers in JavaScript are represented as floating-point values
   - For example: 50, -100, 120.78
2. **Strings of text:**
   - Each string must be enclosed in either:
     - Double quotation " "
     - Single quotation ' '
   - For example: "Marc Jacobs", "My name is "
3. **Boolean value:**
   - The value can either be true or false

JAVASCRIPT: COMMENTS

Comments:

- Will be ignored when the program is executed
- To make a single comment in JavaScript:

```
// This is a single line comment
```

- To make multiple line comments in JavaScript:

```
/*This is a comment
This is also a comment*/
```

JAVASCRIPT: OUTPUT

Output Operation:

- There are different ways to display data in JavaScript:

1. Writing into the HTML output:

- The output will be displayed as HTML output
- The syntax used is:

```
document.write(..... );
```

```
<html>
    <head>
    </head>
    <body>
        <h1>My HTML Web Page</h1>
        <p>I can also include an output using JavaScript.</p>
        <script>
            var message="This is so fun!";
            document.write(message);
        </script>
    </body>
</html>
```

1st way:
JavaScript is placed at the bottom of the body section



JAVASCRIPT: OUTPUT

Output Operation:

- There are different ways to display data in JavaScript:

1. Writing into the HTML output:

```
<html>
    <head>
        <script>
            function DisplayMessage()
            {
                var message="This is so fun!";
                document.write(message);
            }
        </script>

    </head>
    <body>
        <h1>My HTML Web Page</h1>
        <p>I can also include an output using JavaScript.</p>
        <script>
            DisplayMessage();
        </script>
    </body>
</html>
```

2nd way:
JavaScript is placed as a function at the head section



JAVASCRIPT: OUTPUT

Output Operation:

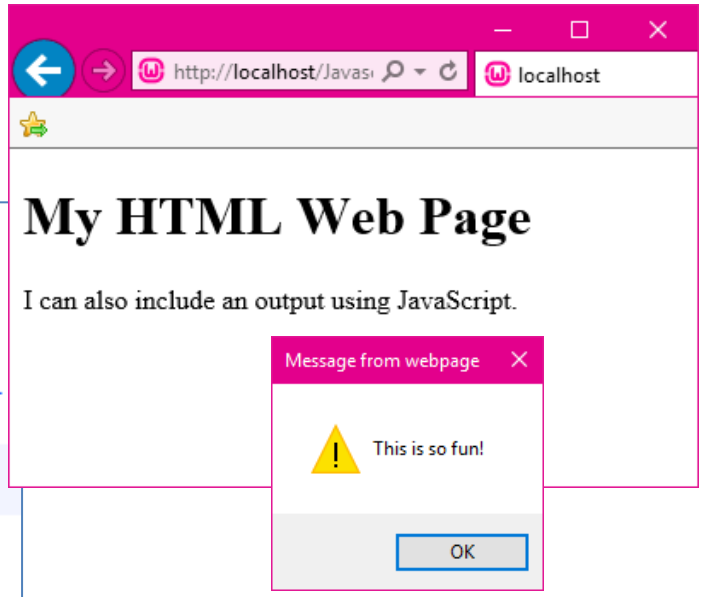
- There are different ways to display data in JavaScript:

2. Writing into an alert box:

- The output will be displayed on a dialogue box which will pop up on the screen
- The syntax used is:

```
alert(.....);
```

```
<html>
    <head>
    </head>
    <body>
        <h1>My HTML Web Page</h1>
        <p>I can also include an output using JavaScript.</p>
        <script>
            var message="This is so fun!";
            alert(message);
        </script>
    </body>
</html>
```



JAVASCRIPT: OUTPUT

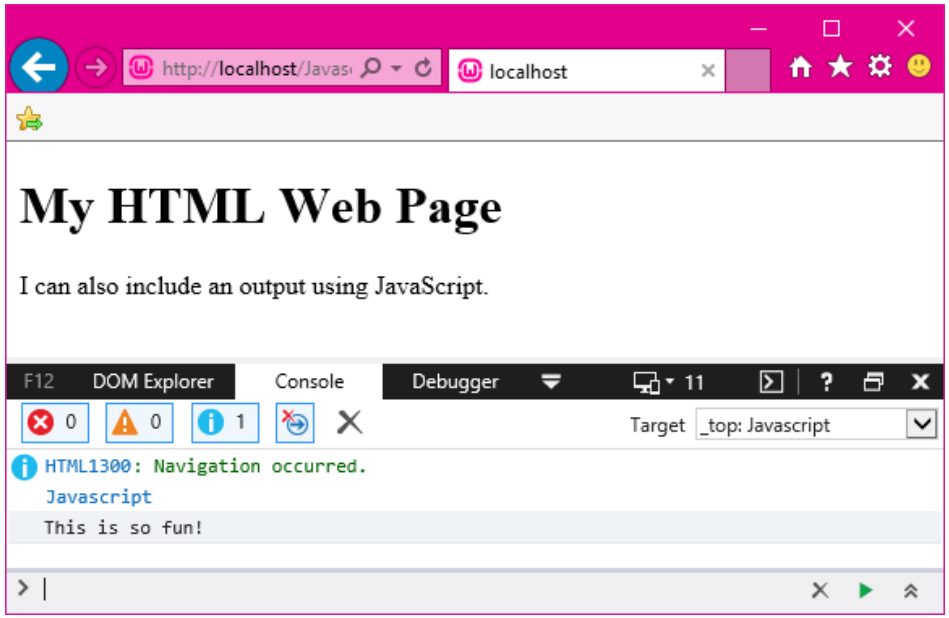
Output Operation:

- There are different ways to display data in JavaScript:

3. Writing into the browser console:

- The output will be displayed on the console of the browser (press F12 on your browser to get to its control)
- The syntax used is: `console.log( ..... );`

```
<html>
    <head>
    </head>
    <body>
        <h1>My HTML Web Page</h1>
        <p>I can also include an output
        using JavaScript.</p>
        <script>
            var message="This is so fun!";
            console.log(message);
        </script>
    </body>
</html>
```



JAVASCRIPT: OUTPUT

Output Operation:

- There are different ways to display data in JavaScript:

4. Writing into an HTML element:

- The output will be displayed as the content of an HTML element
- Will use an ID of the chosen HTML element
- Use this when you want to change the content of an HTML element
- The syntax used is:

```
document.getElementById("htmlElementID").innerHTML = ..... ;
```

```
<html>
    <head>
    </head>
    <body>
        <h1>My HTML Web Page</h1>
        <p>I can also include an output using JavaScript.</p>
        <p id="msg"></p>
        <script>
            var message="This is so fun!";
            document.getElementById("msg").innerHTML = message;
        </script>
    </body>
</html>
```

My HTML Web Page

I can also include an output using JavaScript.

This is so fun!

JAVASCRIPT: OUTPUT

Output Operation:

- There are different ways to display data in JavaScript:

4. Writing into an HTML element:

- Use this when you want **to change the content** of an HTML element

```
<html>
    <head>
    </head>
    <body>
        <h1>My HTML Web Page</h1>
        <p id="change">I can also include an output using JavaScript.</p>
        <p id="msg"></p>
        <script>
            var message1="This is so fun!";
            var message2="The content of this HTML element is changed!";
            document.getElementById("change").innerHTML = message2;
            document.getElementById("msg").innerHTML = message1;
        </script>
    </body>
</html>
```

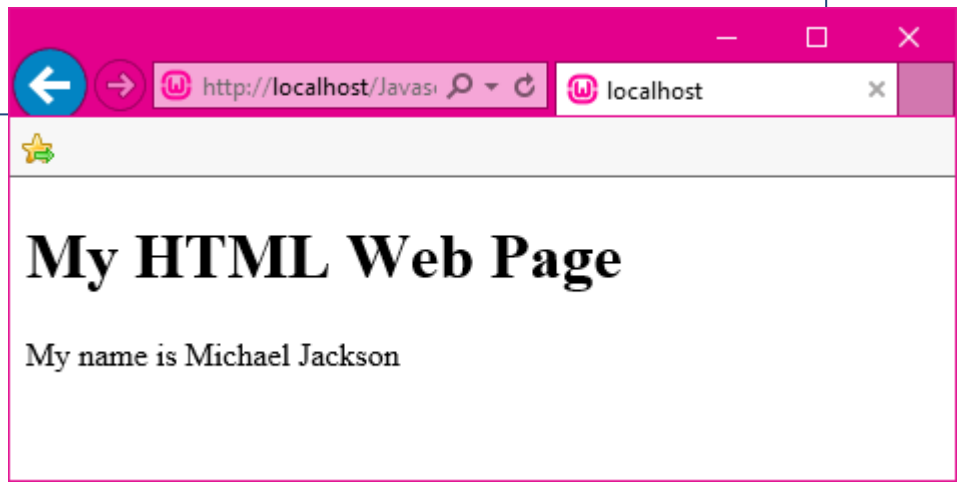


JAVASCRIPT: OUTPUT

Output Operation:

- When you have many data or strings to display:
  You can use "+" to concatenate them

```
<html>
    <head>
    </head>
    <body>
        <h1>My HTML Web Page</h1>
        <script>
            var firstName="Michael";
            var lastName="Jackson";
            document.write("My name is " + firstName + " " + lastName);
        </script>
    </body>
</html>
```



JAVASCRIPT: DIALOG BOX

Dialog Box:

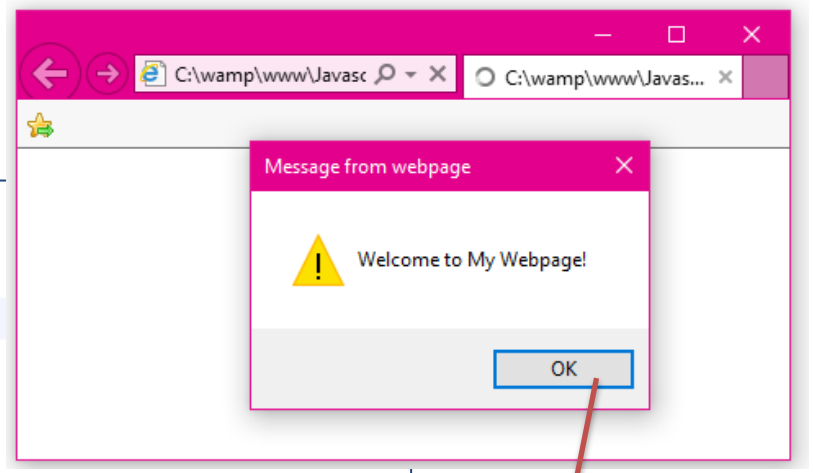
- In JavaScript, there are 3 kinds of pop-up dialog boxes for interacting with the users:

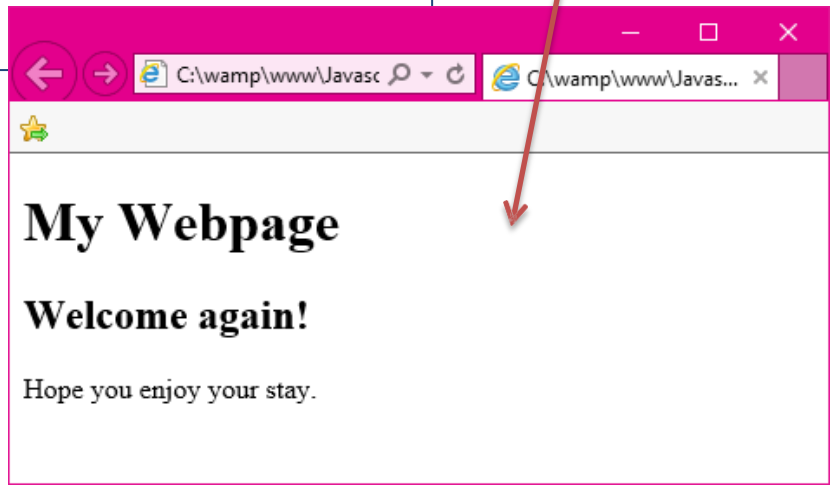
1. **Alert Box:**
   - Will open a dialog box that displays the parameter and an OK button
   - Often used if you want to make sure information comes through to the user:
     Because it will wait for the user until they press the OK button
   - The syntax used is:

```
alert(displayParameter);
```

JAVASCRIPT: DIALOG BOX

```
<html>
    <head>
        <script>
            alert("Welcome to My Webpage!");
        </script>
    </head>
    <body>
        <h1>My Webpage</h1>
        <script>
            document.write("<h2>Welcome again!</h2>");
            document.write("<p>Hope you enjoy your stay.</p>");
        </script>
    </body>
</html>
```





JAVASCRIPT: DIALOG BOX

Dialog Box:

- In JavaScript, there are 3 kinds of pop-up dialog boxes for interacting with the users:

2. Prompt Box:

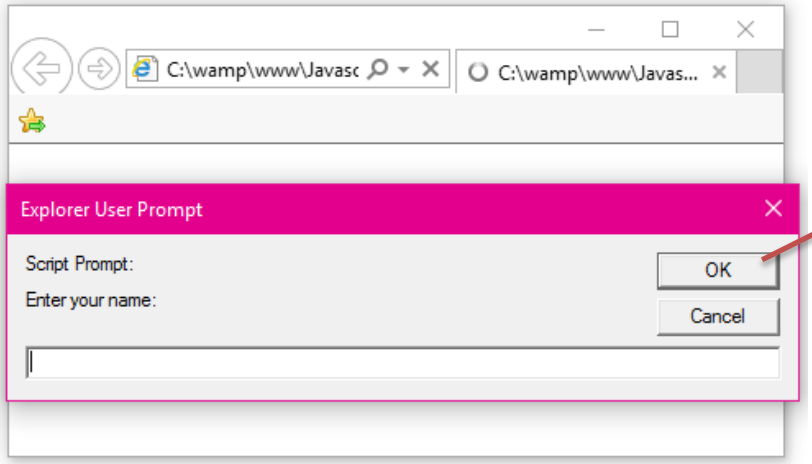
- Will open a dialog box and display the parameter, along with a text box and two buttons (OK and Cancel)
- Often used if you need an input from the user
- The second parameter is for a default response (optional): Will provide an initial value as input
- The syntax used is:

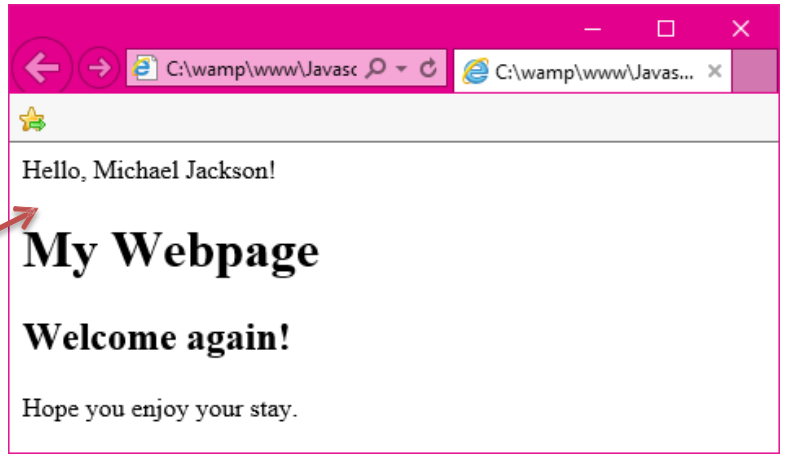
```
prompt(displayParameter,"defaultResponse");
```

JAVASCRIPT: DIALOG BOX

```
<html>
    <head>
        <script>
            var name = prompt("Enter your name: ","");

            document.write("Hello, " + name + "!");
        </script>
    </head>
    <body>
        <h1>My Webpage</h1>
        <script>
            document.write("<h2>Welcome again!</h2>");
            document.write("<p>Hope you enjoy your stay.</p>");
        </script>
    </body>
</html>
```

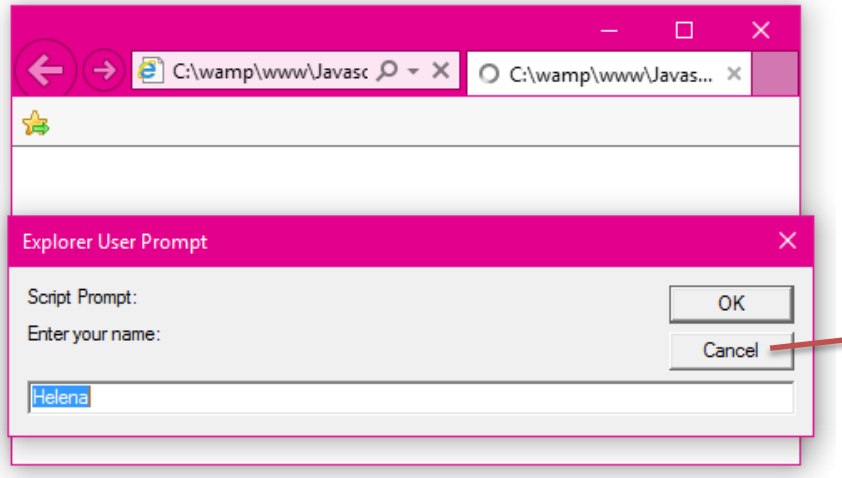


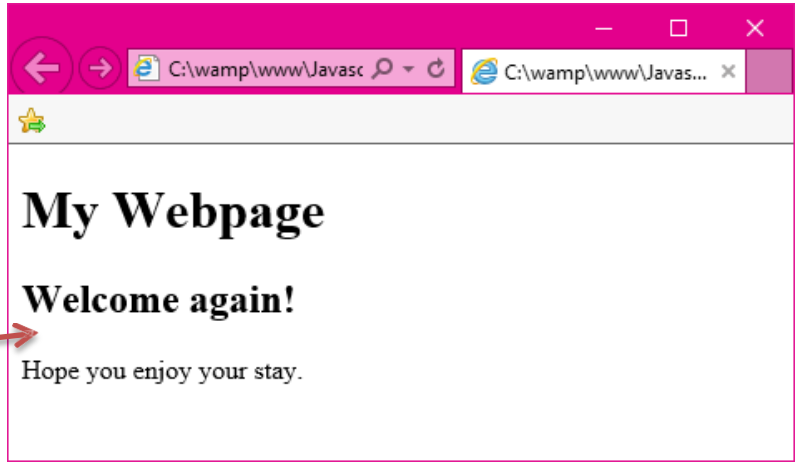


JAVASCRIPT: DIALOG BOX

```
<html>
    <head>
        <script>
            var name = prompt("Enter your name: ", "Helena");

            if(name!=null)
            {
                document.write("Hello, " + name + "!");
            }
        </script>
    </head>
    <body>
        <h1>My Webpage</h1>
        <script>
            document.write("<h2>Welcome again!</h2>");
            document.write("<p>Hope you enjoy your stay.</p>");
        </script>
    </body>
</html>
```





JAVASCRIPT: DIALOG BOX

Dialog Box:

- In JavaScript, there are 3 kinds of pop-up dialog boxes for interacting with the users:

3. Confirm Box:

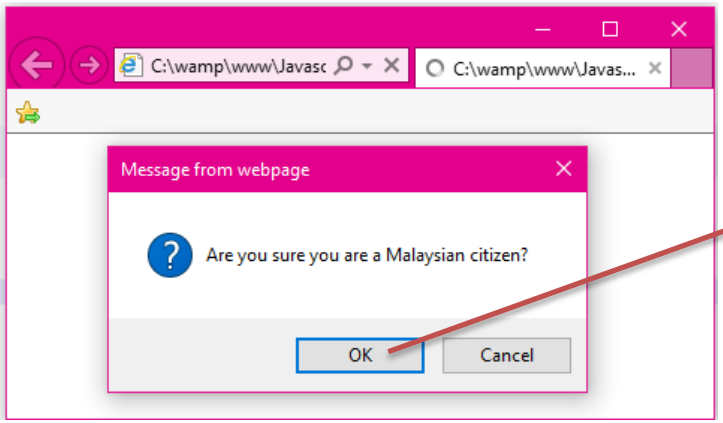
- Will open a dialog box and display the parameter and two buttons (OK and Cancel)
- Will return a Boolean value, depending on which button was pressed (OK: true or CANCEL:false)
- Is often used if you want the user to verify or accept something
- The syntax used is

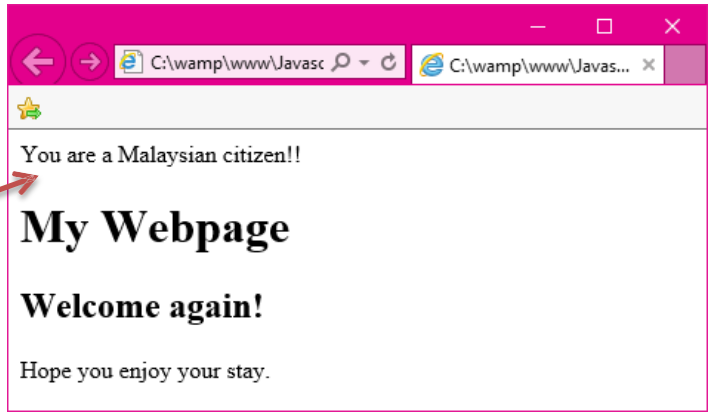
```
confirm(displayParameter);
```

JAVASCRIPT: DIALOG BOX

```
<html>
    <head>
        <script>
            var citizen = confirm("Are you sure you are a Malaysian citizen?");

            if(citizen==true)
            {
                document.write("You are a Malaysian citizen!!");
            }
            else
            {
                document.write("You are not a Malaysian citizen!!");
            }
        </script>
    </head>
    <body>
        <h1>My Webpage</h1>
        <script>
            document.write("<h2>Welcome again!</h2>");
            document.write("<p>Hope you enjoy your stay.</p>");
        </script>
    </body>
</html>
```





JAVASCRIPT: ESCAPE SEQUENCES

Escape Sequences:

- Are used to **display special characters into a string:**
  - What would happen if the following is written?

```
document.write("We must "work hard" to survive");
```



Because strings must be written within quotes, JavaScript might misunderstood the string if the escape sequences is not used

- To display special characters, use **backslash (\)**

```
document.write("We must \"work hard\" to survive");
```

JAVASCRIPT: ESCAPE SEQUENCES

Escape Sequences:

- Below is a list of other special characters that can be added to a text string with the backslash sign (\)

Code	Outputs
\'	single quote
\"	double quote
\&	ampersand
\\	backslash
\n	new line
\r	carriage return
\t	tab
\b	backspace

JAVASCRIPT: PARSING

Parsing User Input:

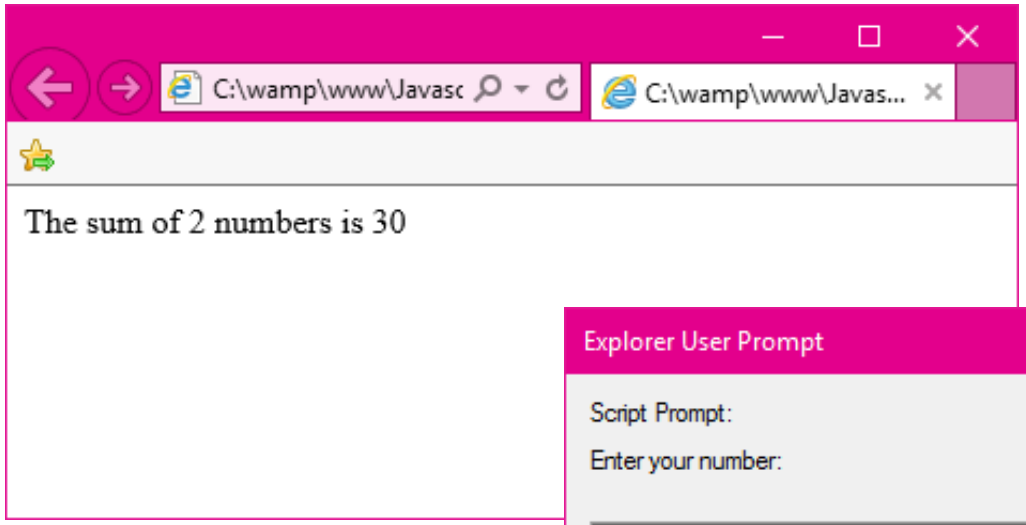
- By default, all input given by users (through prompt function or form) are treated as **string**
- To convert user input into valid number, we must use either one of the following:

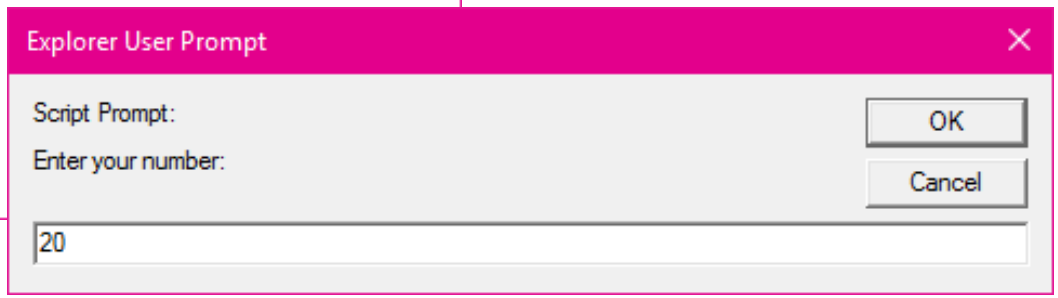
Function	Description
parseFloat()	Convert value into number of type float
parseInt()	Convert value into number of type integer
Number()	Convert value into a number

JAVASCRIPT: PARSING

```
<script>
    var num1 = 10, num2, sum;

    num2 = Number(prompt("Enter your number:", ""));
    sum = num1 + num2;
    document.write("The sum of 2 numbers is " + sum);
</script>
```





JAVASCRIPT: STRING HANDLING

String Handling:

- There are many methods that can be used on a string:

Method	Description
charAt()	Returns the character at the specified index
charCodeAt()	Returns the Unicode of the character at the specified index
concat()	Joins two or more strings, and returns a copy of the joined strings
fromCharCode()	Converts Unicode values to characters
indexOf()	Returns the position of the first found occurrence of a specified value in a string
lastIndexOf()	Returns the position of the last found occurrence of a specified value in a string
match()	Searches for a match between a regular expression and a string, and returns the matches
replace()	Searches for a match between a substring (or regular expression) and a string, and replaces the matched substring with a new substring
search()	Searches for a match between a regular expression and a string, and returns the position of the match
slice()	Extracts a part of a string and returns a new string
split()	Splits a string into an array of substrings
substr()	Extracts the characters from a string, beginning at a specified start position, and through the specified number of character
substring()	Extracts the characters from a string, between two specified indices
toLowerCase()	Converts a string to lowercase letters
toUpperCase()	Converts a string to uppercase letters
valueOf()	Returns the primitive value of a String object

EXAMPLE

```
<script>
    var e = "Hello World";
    document.write(e + "<br>");
    document.write("Length of string: " + e.length + "<br>");
    document.write("Character at location 1: " + e.charAt(1) + "<br>");
    document.write("The substring from location 6 to 9: " + e.substring(6,9) + "<br>");
    document.write("The index of character d: " + e.indexOf("d") + "<br>");
    document.write("The index of substring World: " + e.indexOf('World') + "<br>");
    document.write("Capitalize string: " + e.toUpperCase() + "<br>");
    document.write("Decapitalize string: " + e.toLowerCase() + "<br>");
</script>
```

Hello World
Length of string: 11
Character at location 1: e
The substring from location 6 to 9: Wor
The index of character d: 10
The index of substring World: 6
Capitalize string: HELLO WORLD
Decapitalize string: hello world

JAVASCRIPT: NONSTANDARD METHOD

Non-standard Method:

- Below are some JavaScript method that can also be used:
  - Instead of using HTML code for bold:

```
.bold()
```

  - Instead of using HTML code for font colour:

```
.fontcolor(" . ")
```

  - Instead of using for links:

```
.link(" . ")
```

EXAMPLE

```
<script>
    var e = "Hello World";

    //<b>Hello World</b>
    document.write(e.bold());
    document.write("<br>");

    //<font color="red">Hello World</font>
    document.write(e.fontcolor("red"));
    document.write("<br>");

    //<a href="index.html">Click Here</a>
    document.write("Click Here".link("index.html"));
</script>
```

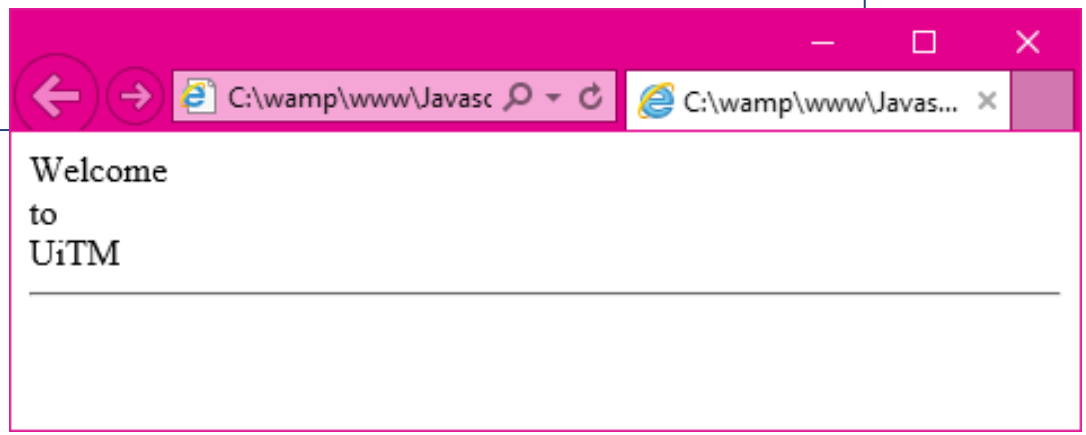
Hello World
Hello World
Click Here

JAVASCRIPT: HTML CODE

HTML Code:

- Can be included in JavaScript (by writing it as HTML output)
- However, it must be enclosed inside a quotation mark (“ ”or‘ ’)

```
<html>
    <body>
        <script>
            document.write("Welcome<br>to<br> UiTM<hr>");
        </script>
    </body>
</html>
```



JAVASCRIPT: HTML CODE

```
<html>
    <head>
        <script>
            var colour;
            colour="lightblue";
            document.write("<body bgcolor=" + colour + ">");
        </script>
    </head>
    The background of this page is :
    <script>
        document.write(colour);
    </script>
    </body>
</html>
```

